Variation of Parameters

ANSWER KEY

The Wronskian

1 For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
$y_1' = e^x$, $y_2' = 2e^{2x}$. $W = e^x(2e^{2x}) - e^x(e^{2x}) = 2e^{3x} - e^{3x} = e^{3x}$.

2 For $y_1 = \cos x$, $y_2 = \sin x$, compute W.
$y_1' = -\sin x$, $y_2' = \cos x$. $W = \cos x \cos x - (-\sin x)\sin x = \cos^2 x + \sin^2 x = 1$.

3 For $y_1 = e^{3x}$, $y_2 = xe^{3x}$ (solutions of a repeated-root equation), compute W.
$y_1' = 3e^{3x}$, $y_2' = e^{3x} + 3xe^{3x}$. $W = e^{3x}(e^{3x} + 3xe^{3x}) - 3e^{3x}(xe^{3x}) = e^{6x} + 3xe^{6x} - 3xe^{6x} = e^{6x}$.

The variation of parameters formula

4 State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.
$y_p = -y_1 \int \frac{y_2\, g}{W}\, dx + y_2 \int \frac{y_1\, g}{W}\, dx$.

5 Explain why the ODE must first be written in standard form $y'' + p(x)y' + q(x)y = g(x)$ (leading coefficient 1) before applying the variation of parameters formula.

The formula's Wronskian and integrals are derived under the assumption that $g(x)$ is exactly the right side once the leading coefficient of y'' is 1. If the equation is given as $a(x)y'' + b(x)y' + c(x)y = f(x)$ with $a(x) \neq 1$, using $f(x)$ directly in place of $g(x)$ would introduce an extra factor of $a(x)$ and give a wrong particular solution — the equation must first be divided by $a(x)$, giving $g(x) = f(x)/a(x)$.

A forcing term with no undetermined-coefficients trial form

6 Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.
Here $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \tan x$. Since $\int \cos x \tan x\, dx = \int \sin x\, dx = -\cos x$ and $\int \sin x \tan x\, dx = \int \sec x\, dx - \int \cos x\, dx = \ln|\sec x + \tan x| - \sin x$, we get $y_p = -\cos x[\ln|\sec x + \tan x| - \sin x] + \sin x(-\cos x) = -\cos x \ln|\sec x + \tan x|$. General solution: $y = C_1 \cos x + C_2 \sin x - \cos x \ln|\sec x + \tan x|$.

7 Solve $y'' + y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.
$y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \sec x$. $\frac{y_2 g}{W} = \frac{\sin x}{\cos x} = \tan x$, so $\int = -\ln|\cos x|$. $\frac{y_1 g}{W} = \frac{\cos x}{\cos x} = 1$, so $\int = x$. Then $y_p = -\cos x(-\ln|\cos x|) + \sin x(x) = \cos x \ln|\cos x| + x \sin x$. General solution: $y = C_1 \cos x + C_2 \sin x + \cos x \ln|\cos x| + x \sin x$.

A repeated root, with forcing dividing by x

8 Solve $y'' - 2y' + y = \dfrac{e^x}{x}$ $(x > 0)$ using variation of parameters.

$r^2 - 2r + 1 = (r-1)^2 = 0$, so $y_1 = e^x$, $y_2 = xe^x$, and $W = e^{2x}$. With $g = e^x/x$: $\dfrac{y_2 g}{W} = 1$ so $\int = x$; $\dfrac{y_1 g}{W} = \dfrac{1}{x}$ so $\int = \ln x$. Then $y_p = -e^x(x) + xe^x(\ln x) = xe^x(\ln x - 1)$. Absorbing the $-xe^x$ term into the arbitrary constant on y_2, the general solution is $y = C_1 e^x + C_2 xe^x + xe^x \ln x$.

9 Solve $y'' - 4y' + 4y = \dfrac{e^{2x}}{x^2}$ $(x > 0)$ using variation of parameters.

$r^2 - 4r + 4 = (r-2)^2 = 0$, so $y_1 = e^{2x}$, $y_2 = xe^{2x}$, $W = e^{4x}$. With $g = e^{2x}/x^2$: $\dfrac{y_2 g}{W} = \dfrac{1}{x}$, integral $= \ln x$; $\dfrac{y_1 g}{W} = \dfrac{1}{x^2}$, integral $= -\dfrac{1}{x}$. Then $y_p = -e^{2x}\ln x + xe^{2x}\left(-\dfrac{1}{x}\right) = -e^{2x}\ln x - e^{2x}$. Absorbing the constant $-e^{2x}$ term into C_1, the general solution is $y = C_1 e^{2x} + C_2 xe^{2x} - e^{2x}\ln x$.

Comparing with undetermined coefficients

10 Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. With $g = e^{2x}$: $\dfrac{y_2 g}{W} = -\frac{1}{2}e^x$, integral $-\frac{1}{2}e^x$; $\dfrac{y_1 g}{W} = -\frac{1}{2}e^{3x}$, integral $-\frac{1}{6}e^{3x}$. So $y_p = -e^x(-\frac{1}{2}e^x) + e^{-x}(-\frac{1}{6}e^{3x}) = \frac{1}{2}e^{2x} - \frac{1}{6}e^{2x} = \frac{1}{3}e^{2x}$. Check by undetermined coefficients: try Ae^{2x} (no resonance, since 2 is not a root of $r^2 - 1 = 0$): $4A - A = 1 \Rightarrow A = \frac{1}{3}$. Matches. General solution: $y = C_1 e^x + C_2 e^{-x} + \frac{1}{3}e^{2x}$.

11 Solve $y'' - y = 4$ (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess $y_p = A$.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$, $g = 4$. $\dfrac{y_2 g}{W} = -2e^{-x}$, integral $2e^{-x}$; $\dfrac{y_1 g}{W} = -2e^x$, integral $-2e^x$. So $y_p = -e^x(2e^{-x}) + e^{-x}(-2e^x) = -2 - 2 = -4$. Undetermined coefficients: try $y_p = A$, giving $-A = 4 \Rightarrow A = -4$. Matches. General solution: $y = C_1 e^x + C_2 e^{-x} - 4$.

When to prefer each method

12 For $y'' + 4y = x^3 - 2x$, which method is more efficient: undetermined coefficients or variation of parameters? Explain.

Undetermined coefficients is far more efficient here: the forcing is a plain polynomial (no resonance issues), so a simple trial $y_p = Ax^3 + Bx^2 + Cx + D$ solves it by matching coefficients — no integration required. Variation of parameters would work too, but requires computing W and two integrals unnecessarily.

13 For $y'' + y = \csc x$ (for $0 < x < \pi$), which method must be used, and why?

Variation of parameters is required. $\csc x = 1/\sin x$ is not a polynomial, exponential, sine/cosine, or a finite product of these, so undetermined coefficients has no valid trial form; only variation of parameters, which places no restriction on the forcing term, can produce a particular solution.

Solving $y'' + y = \csc x$ in full

14 Complete the solution of $y''+y=\csc x$ ($0<x<\pi$) started above, using $y_1=\cos x$, $y_2=\sin x$, $W=1$.
$\frac{y_2 g}{W}=\frac{\sin x}{\sin x}=1$, so $\int=x$. $\frac{y_1 g}{W}=\frac{\cos x}{\sin x}=\cot x$, so $\int=\ln|\sin x|$. Then $y_p=-\cos x(x)+\sin x\ln|\sin x|=-x\cos x+\sin x\ln|\sin x|$. General solution: $y=C_1\cos x+C_2\sin x-x\cos x+\sin x\ln|\sin x|$.

15 Verify (by checking the coefficient of the leading term only) that $y_p=-x\cos x$ contributes a term of the correct order when substituted into $y''+y$.

For $y_p=-x\cos x$: $y_p'=-\cos x+x\sin x$, $y_p''=\sin x+\sin x+x\cos x=2\sin x+x\cos x$. So $y_p''+y_p=2\sin x+x\cos x-x\cos x=2\sin x$ — this piece alone contributes $2\sin x$, which combines with the contribution from $\sin x\ln|\sin x|$ to produce the required $\csc x$ on the right side (a full check of the second piece is a longer computation, but this confirms the $-x\cos x$ term is doing real work, not just decoration).

Variation of parameters with a shifted independent variable

16 For the Cauchy–Euler-type substitution used in some ODE courses, $y_1=x$, $y_2=x^{-1}$ solve $x^2y''+xy'-y=0$ for $x>0$. Rewrite this equation in the standard form $y''+p(x)y'+q(x)y=g(x)$ needed for variation of parameters when the right side is $x^2\cdot 3x=3x^3$ (i.e., solve $x^2y''+xy'-y=3x^3$).
Dividing by x^2: $y''+\frac{1}{x}y'-\frac{1}{x^2}y=3x$, so $g(x)=3x$ (NOT $3x^3$ — the division by the leading coefficient x^2 is exactly the step the earlier question about standard form warns about).

17 Continue the problem above: with $y_1=x$, $y_2=x^{-1}$, compute $W=y_1y_2'-y_1'y_2$.
$y_1'=1$, $y_2'=-x^{-2}$. $W=x(-x^{-2})-1(x^{-1})=-x^{-1}-x^{-1}=-2x^{-1}$.

18 Finish solving $x^2y''+xy'-y=3x^3$ ($x>0$) using $y_1=x$, $y_2=x^{-1}$, $W=-2x^{-1}$, and $g(x)=3x$ (from the two questions above).
$\frac{y_2 g}{W}=\frac{x^{-1}(3x)}{-2x^{-1}}=-\frac{3x}{2}$, so $\int=-\frac{3x^2}{4}$. $\frac{y_1 g}{W}=\frac{x(3x)}{-2x^{-1}}=-\frac{3x^3}{2}$, so $\int=-\frac{3x^4}{8}$. Then $y_p=-x\left(-\frac{3x^2}{4}\right)+x^{-1}\left(-\frac{3x^4}{8}\right)=\frac{3x^3}{4}-\frac{3x^3}{8}=\frac{3x^3}{8}$. General solution: $y=C_1x+C_2x^{-1}+\frac{3x^3}{8}$.

19 Check the particular solution $y_p=\frac{3}{8}x^3$ found above by substituting directly into $x^2y''+xy'-y=3x^3$.
$y_p'=\frac{9}{8}x^2$, $y_p''=\frac{9}{4}x$. Then
$x^2y_p''+xy_p'-y_p=x^2\left(\frac{9}{4}x\right)+x\left(\frac{9}{8}x^2\right)-\frac{3}{8}x^3=\frac{9}{4}x^3+\frac{9}{8}x^3-\frac{3}{8}x^3=\frac{18}{8}x^3+\frac{9}{8}x^3-\frac{3}{8}x^3=\frac{24}{8}x^3=3x^3$.
Matches.

A summary comparison

20 List two advantages of variation of parameters over undetermined coefficients.

(1) It works for ANY continuous forcing term $g(x)$, not just polynomials/exponentials/sines-cosines and their products — it handles $\tan x$, $\sec x$, $1/x$, $\ln x$, and more. (2) It gives a single unified formula/procedure, with no need to guess a trial form or check for resonance case-by-case.

21 List one advantage of undetermined coefficients over variation of parameters, when it applies.

It is purely algebraic — matching coefficients — with no integration required, so it is typically much faster when the forcing term is one of its supported families (and the resulting integrals in variation of parameters can themselves be difficult or impossible to evaluate in closed form).

22 True or false, with a one-sentence justification: "Variation of parameters requires knowing the general solution of the associated homogeneous equation."

True — the formula uses two independent homogeneous solutions y_1, y_2 and their Wronskian W directly, so the homogeneous equation must be solved first, exactly as with undetermined coefficients (which also needs y_c to check for resonance).

23 For $y'' + 4y = 8\tan 2x$ ($-\frac{\pi}{4} < x < \frac{\pi}{4}$), which method must be used, and what are y_1, y_2, and W?

Variation of parameters is required, since $\tan 2x$ is outside the undetermined-coefficients family. From $r^2 + 4 = 0$: $y_1 = \cos 2x$, $y_2 = \sin 2x$.
$W = y_1 y_2' - y_1' y_2 = \cos 2x(2\cos 2x) - (-2\sin 2x)\sin 2x = 2\cos^2 2x + 2\sin^2 2x = 2$.

24 In one sentence, summarize when a student should reach for variation of parameters instead of undetermined coefficients.

Reach for variation of parameters whenever the forcing term $g(x)$ is not a polynomial, exponential, sine/cosine, or product of these (e.g., it involves $\tan$, $\sec$, $\csc$, $\ln$, or a quotient like $1/x$) — otherwise undetermined coefficients is faster.